THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 0	every point on the x-axis has $y = 0$
2	**(c)** $(-5,\ 7)$	reflection in the y-axis changes the sign of the x-coordinate
3	**(a)** $(-7,\ 2)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
4	**(d)** -9	$2(-6) + (3) = -9$
5	**(a)** positive slope	values increase with x
6	**(c)** $\frac{3}{11}$	standard method ⇒ $\frac{3}{11}$
7	**(a)** terminating	denominator 20 is of the form 2^a5^b
8	**(c)** $\frac{1}{7}$	7 is a prime other than 2, 5
9	**(b)** $25x^2 + 20x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
10	**(c)** $(x-7)\,(x+6)$	find a, b with $a + b$ = middle coefficient and ab = constant
11	**(b)** 115°	opposite angles of a cyclic quadrilateral add to 180°
12	**(b)** bisects the chord	perpendicular from the centre bisects the chord
13	**(d)** exactly one	unique circumcircle
14	**(c)** the same for all circles (π)	the C/D ratio is the constant π
15	**(d)** 14 cm	$r = \frac{C}{2\pi} = \frac{88\times7}{44} = 14$
16	**(d)** 6 cm^2	$s = 6$, area $= \sqrt{6\cdot3\cdot2\cdot1} = 6$
17	**(c)** $\frac{2}{11}$	2 favourable out of 11 letters
18	**(a)** 0.65	$1 - 0.35$
19	**(a)** 29	1, 5, 13, 29
20	**(a)** $n = 122$	$5n - 3 = 607 \Rightarrow n = 122$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$O(0,0),\ A(7,0),\ B(7,4),\ C(0,4)$	2
22	$25 + 15 \times 8 = ₹145$.	1+1
23	$\left\lvert -\frac{1}{2} - \frac{5}{2} \right\rvert = 3$.	1+1
24	$119^2 = (100+10+9)^2 = 10000 + 100 + 81 + 2000 + 1800 + 180 = 14161$.	2
25	$d = \sqrt{13^2 - 5^2} = 12$ cm for each; equal chords are equidistant from the centre.	1+1
26	A median divides a triangle into two triangles of equal area (equal bases, same height): $\text{ar}(ABD) = 24$ cm^2.	1+1
27	(i) $\frac{3}{20}$. (ii) $\frac{17}{20}$.	1+1
28	Stage n has 3^n black triangles: Stage 4 has 81 and Stage 5 has 243.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Diagonals bisect each other: midpoint of $AC = (3.5, 4)$. If $D = (x, y)$ then $\frac{4+x}{2} = 3.5 \Rightarrow x = 3$, $\frac{3+y}{2} = 4 \Rightarrow y = 5$. $D = (3, 5)$	1+1+1
30	$V = 77h$: 385, 693, 1001 cm^3. Consecutive differences are equal (308) — a linear pattern.	2+1
31	LHS $= \frac{1}{2} \times \frac{7}{12} = \frac{7}{24}$; RHS $= \frac{1}{8} + \frac{1}{6} = \frac{3}{24} + \frac{4}{24} = \frac{7}{24}$ ✓	1+1+1
32	$r^2 - r - 42 = (r-7)(r+6)$; $x^2 + 7x + 12 = (x+3)(x+4)$.	1½+1½
33	Area $= \frac{1}{2}(12+20) \times 7 = 112$ cm^2.	2+1
34	April, June, September, November ⇒ $\frac{4}{12} = \frac{1}{3}$ (3).	3
35	The angle at the origin is 90°, so the hypotenuse joining $(6,0)$ and $(0,8)$ is a diameter (1). Its midpoint $(3,4)$ is the centre (1); radius $= \frac{1}{2}\sqrt{36+64} = 5$ (1).	1+1+1

	OR: Let $AB = CD$. In $\triangle AOB$ and $\triangle COD$: $OA = OC$, $OB = OD$, $AB = CD$ ⇒ congruent (SSS) (2). So $\angle AOB = \angle COD$ (1).	2+1
36	$1, 2, 3, 5, 8, 13, 21, 34, 55, 89.$	3
	OR: $3,\ 6,\ 30,\ 870.$	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\lvert -4-3\rvert = 7$; $\left\lvert -\frac{5}{2} - \frac{7}{2}\right\rvert = 6$. (ii) $\frac{7}{9}$, 0, $\frac{5}{3}$. (iii) $\frac{13}{4} = 3.25$ lies between 3 and 4.	2+1+2
	OR: $x = 0.\overline{142857}$, $10^6 x = 142857.\overline{142857}$, $999999x = 142857 \Rightarrow x = \frac{142857}{999999} = \frac{1}{7}$ (3). $2 \times 142857 = 285714$, $3 \times 142857 = 428571$ — same digits cycled (2).	3+2
38	(i) 80° (1). (ii) $OB = OC$ ⇒ $\angle OBC = 50^\circ$ (1). (iii) Same segment ⇒ $\angle BDC = 40^\circ$ (1). (iv) Half-chord $= \sqrt{60^2 - 36^2} = 48$; $BC = 96$ cm (2).	1+1+1+2
	OR: (i) Diagonal $= \sqrt{25 + 144} = 13$; angle in a semicircle ⇒ it is a diameter, radius 6.5 (2). (ii) $2 \times \frac{1}{2} \times 5 \times 12 = 60$ sq. units (2). (iii) Its sides in order are 5,5,12,12 (a kite) — opposite sides are not equal, so it is not a rectangle (1).	2+2+1
39	(i) If the diameters are $d_1, \ldots, d_n$ then $d_1 + \cdots + d_n = PQ$; length $= \pi(d_1 + \cdots + d_n)/2 = \pi PQ/2$, the same as the big semicircle (3). (ii) $\frac{22}{7} \times \frac{28}{2} = 44$ cm; also $\frac{22}{7} \times \frac{7+7+14}{2} = 44$ cm ✓ (2).	3+2
	OR: (i) $1 \times 19 = 19$, $2 \times 18 = 36$, $3 \times 17 = 51$, $4 \times 16 = 64$, $5 \times 15 = 75$, $6 \times 14 = 84$, $7 \times 13 = 91$, $8 \times 12 = 96$, $9 \times 11 = 99$, $10 \times 10 = 100$ (3). (ii) Greatest 100 (10×10), least 19 (1×19) (1). (iii) It is a square (1).	3+1+1
40	(i) $S_{58} - S_{24} = 29 \times 59 - 12 \times 25 = 1711 - 300 = 1411$ (2). (ii) 34 terms; $\frac{25+58}{2} \times 34 = 1411$ ✓ (2). (iii) Āryabhaṭa, in the Āryabhaṭīya, Chapter 2, verse 19 (1).	2+2+1
	OR: (i) $-5, -2, 1, 4, 7$ (1). (ii) $t_n = 3n - 8$ (1). (iii) $3n - 8 = 52 \Rightarrow n = 20$, yes (1). (iv) $\frac{20}{2}(-5 + 52) = 470$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER